

Valeria Mascolo

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About

Research scientist at the University of Reading specializing in *climate tipping points and extremes*. **PhD in Physics** from École Normale Supérieure de Lyon as **Marie Skłodowska-Curie ITN H2020 fellow**. **Organiser of the RMetS Weather and Climate Conference 2026**. **Chief blog editor** and events organiser for EGU's *Climate: Past, Present & Future* and *Nonlinear Processes in Geosciences* ECS divisions. **Member of Risk-KAN and COST Action ANTICIPATE** networks, connecting research with risk governance and early-warning systems. **Finalist** of the [French Complex Systems Society PhD Award 2025](#). **Passionate** about translating climate science into actionable knowledge that addresses the societal, economic, environmental, and health impacts of climate change. Outside research: art history enthusiast, baker, swimmer, and runner.

Professional experience

- since 2025 – **Research Scientist, National Centre for Atmospheric Science, University of Reading**
Simulating, forecasting and associated impacts of tipping points: Subpolar Gyre and Greenland Ice Sheet, as part of PROMOTE, advised by Prof. Reinhard Schiemann and Dr. Andrea Dittus.
- 2020–2021 **Trainee at the European Commission's Joint Research Centre, Text Mining & Analysis Competence Centre team**, Machine learning to analyze social media sentiment on climate engagement, focusing on the European Green Deal.

Professional service & leadership

- since 2026 – **Steering Committee Member, Risk-KAN**, Knowledge Action Network on Emergent Risks and Extreme Events - Reducing Disaster Risks under Environmental Change
- since 2026 – **Working groups Member, COST Action ANTICIPATE: extended-range multi-hazard predictions and early warnings**, European research and innovation network funded as a COST Action CA24144 ANTICIPATE under the European Cooperation in Science and Technology framework.
- 2026 **Organizing Committee Member, Royal Meteorological Society, RMets Weather and Climate Conference 2026**, University of Exeter
- 2026 **Co-organiser, workshop ExREC, EXtremes and Rare Events in Climate and related applications**, University of Reading
- since 2025 – **Reviewer for Weather and Climate Dynamics and International Journal of Climatology journals**

Education

- 2021–2024 **Ph.D. in Physics, Ecole Normale Supérieure de Lyon**
Title: *Studying extreme heatwaves using novel theoretical approaches*, advised by Prof. Freddy Bouchet. **Marie Skłodowska-Curie ITN H2020 fellow**, within project EDIPI. Published manuscript available [here](#).
- 2019–2020 **M.Sc. in Complex systems modelling, King's College London, Highest honors**
Dissertation: *Analysis of Bayesian classification - Naive Bayes approach for the analysis of medical data*
- 2016–2019 **B.Sc. in Physics Engineering, Politecnico di Torino, Highest honors**
Dissertation: *Deterministic and stochastic analysis of epidemics. SIS and SIR approaches to model epidemics.*

Outreach and volunteering

- since 2025 – **EGU's Climate: Past, Present & Future and Nonlinear processes in Geosciences ECS division member**, Co-chief blog editor, events and meetings organiser.

since 2025 – **Climate Ambassador**, Member of [Climate Ambassadors](#), to support nurseries, schools and colleges to build their climate action plans.

Research visits

- Oct. – Dec. 2023 **Visiting PhD student of Prof. Francesco Ragone**, *Royal Meteorological Institute of Belgium*
- Oct. – Dec. 2022 **Visiting PhD student of Prof. Nili Harnik**, *Tel Aviv University, Department of Geoscience*

Teaching

- 2026 **Teaching assistant, *Climate Change***, MSc. *Meteorology, Department of Meteorology, University of Reading*

Schools and trainings

2025– **Impacts of climate change on health courses**

- South and Southeast Asia Climate and Health Responders, Columbia University.
- European Climate Resilient Health Systems Course, Columbia University.

2021–2024 **Third academic year (2023-2024)**

- EDIPI ITN 5th training week at KIT, Karlsruhe, Germany
- EDIPI ITN 6th training week - *Network-wide ESR Research and Soft Skills Training* at Cyprus Institute, Nicosia, Cyprus
- Participation to the European Geosciences Union General Assembly, Wien, Austria
- Workshop - *Prévisibilité dans les sciences de l'atmosphère et Points de bascule* for the Group De Recherche "Défis théoriques pour les sciences du climat", Institut Henri Poincaré, Paris, France
- Workshop - *Interfaces dans le système climatique*, for the Group De Recherche "Défis théoriques pour les sciences du climat", Grenoble, France

Second academic year (2022-2023)

- **Workshop** - *Mathematics and theoretical physics for climate dynamics*, Ecole Normale Supérieure de Lyon, Lyon, France
- **Summer school** - *Climate sensibility*, Ecole Normale Supérieure de Lyon, Lyon, France
- EDIPI ITN 4th training week - *Idealized modelling in climate sciences*, Institut Royal Météorologique, Brussels, Belgium
- **Summer school** - *Out of equilibrium statistical physics*, Beg Rohu, France
- Course - *Granular, jammed and disordered media*, Ecole Normale Supérieure de Lyon, Lyon, France
- Participation to the European Geosciences Union General Assembly, Wien, Austria

First academic year (2021-2022)

- Course - *Large deviations in physics*, Ecole Normale Supérieure de Lyon, Lyon, France
- Course - *Climate change, climate change impacts, and energetic transition*, Ecole Normale Supérieure de Lyon, Lyon, France
- **Summer school** - *Artificial intelligence for detection and attribution of climate extremes*, XAIDA project, ICTP, Trieste, Italy
- **School** - *Machine Learning and the Physics of Climate* and Workshop on *Extreme Events in Weather and Climate*, Fletcher Hotel, Berg en Daal, The Netherlands
- EDIPI ITN 1st & 2nd training weeks - *Basics of statistics, probability, extreme events modelling and extreme value theory*, Uppsala University, Uppsala, Sweden
- EDIPI ITN 3rd training week - *Attribution studies, tipping point of climate and advanced statistical methods for extremes*, Ecole Normale Supérieure, Paris, France
- Journées de la physique statistique, Ecole Normale Supérieure, Paris, France
- Journées du lancement du Group De Recherche 'Défis théoriques pour les sciences du climat', Auditorium TOTEM, Paris, France

Talks and posters

- 2026 **VU Amsterdam**, Workshop on "Dynamics of Rossby Waves, Compound Extremes and Their Impacts". Presentation of the work: Rare events algorithm study of extreme double jet summers and their connection to heatwaves over Eurasia.

- 2025 **University of Reading**, PROMOTE kickoff meeting. Presentation of the work: Rare events algorithm study of extreme double jet summers and their connection to heatwaves over Eurasia.
- 2025 **EMS2025**, Presentation of the work: Rare events algorithm study of extreme double jet summers and their connection to heatwaves over Eurasia.
- 2025 **University of Chicago**, Invited speaker. Presentation of the work: Gaussian framework and optimal projection of weather fields for prediction of extreme events.
- 2025 **Stormtracks2025**, Poster presentation of the work: Rare events algorithm study of extreme double jet summers and their connection to heatwaves over Eurasia
- 2025 **ETH Zurich**, Invited speaker. Presentation of the work: Gaussian framework and optimal projection of weather fields for prediction of extreme events. Invited by Prof. Dr. Daniela Domeisen
- 2024 **European Geosciences Union General Assembly, EGU24**, Presentation of the work: A Gaussian Framework for Optimal Prediction of Extreme Heatwaves
- 2023 **GdR Defi théoriques pour les sciences du climat**, Poster presentation of the work: Heatwaves prediction using Gaussian assumption
- 2023 **Beg Rohu School of Out of equilibrium statistical physics**, Presentation of the work: Heatwaves prediction using Gaussian assumption
- 2023 **European Geosciences Union General Assembly, EGU23**, Presentation of the work: Comparing the influence of Atlantic Multidecadal Variability and spring soil moisture on European summer heatwaves
- 2022 **Tel-Aviv University, Department of Geoscience**, Presentation of the work: Comparing the influence of Atlantic Multidecadal Variability and spring soil moisture on European summer heatwaves
- 2022 **GdR Defi théoriques pour les sciences du climat**, Poster presentation of the work: Comparing the influence of Atlantic Multidecadal Variability and spring soil moisture on European summer heatwaves

Skills

Programming Python, MATLAB, C, Bash, $\text{\LaTeX}$, Microsoft Office

Languages Italian, English, French, Neapolitan

Publications

- [1] Valeria Mascolo. *Studying extreme heatwaves using novel theoretical approaches*. PhD thesis, École normale supérieure de Lyon, 2024. <https://theses.hal.science/tel-04843914/>.
- [2] Valeria Mascolo, Clément Le Priol, Fabio D’Andrea, and Freddy Bouchet. Comparing the influence of atlantic multidecadal variability and spring soil moisture on european summer heat waves. *Oxford Open Climate Change*, 5(1), 2025. <https://doi.org/10.1093/oxfclm/kgae023>.
- [3] Valeria Mascolo, Alessandro Lovo, Corentin Herbert, and Freddy Bouchet. Gaussian framework and optimal projection of weather fields for prediction of extreme events. *Journal of Advances in Modeling Earth Systems*, 17(6), 2025. <https://doi.org/10.1029/2024MS004487>.
- [4] Valeria Mascolo, Francesco Ragone, Nili Harnik, and Freddy Bouchet. Rare events algorithm study of extreme double jet summers and their connection to heatwaves over Eurasia, 2025. <https://arxiv.org/abs/2507.13101>, Submitted to *Journal of Climate*.

References

Prof. Reinhard Schiemann

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Dr. Andrea Dittus

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Prof. Freddy Bouchet

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Prof. Nili Harnik

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Prof. Corentin Herbert

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Prof. Francesco Ragone

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